

PulseNet Customer Churn Analytics

End-to-end Data Analyst Portfolio Project

Python + SQL + Tableau

Project type: Fictional customer/subscription analytics case study

Objective: Investigate customer churn, understand important associated patterns, quantify customer/revenue exposure, and communicate findings through Tableau dashboards.

Metric	Validated value
Total customers	12,000
Active customers	10,394
Churned customers	1,606
Overall churn rate	13.38%
Cleaned payment records	117,501
Successful payments	108,402
Failed payments	9,099
Churn period	March–June 2026
Churn peak	May 2026 — 477 customers
Historical successful payment value associated with churned customers	\$23.72M

1. Business Context

PulseNet is a fictional subscription/customer-service business. Leadership wants to understand who is leaving, which customer segments show higher churn, where churn is concentrated, and what customer or revenue exposure may deserve attention.

The project follows an end-to-end analyst workflow: data understanding → cleaning and validation → Python analysis → SQL analysis → Tableau dashboards → QA → business recommendations.

2. Dataset & Data Model

The project contains five related tables. **customers** is the central customer-level table. The remaining four tables contain repeated Customer_ID values and represent one-to-many customer activity.

Table	Primary key	Purpose
customers	Customer_ID	Customer profile, acquisition, status and churn date
payments	Payment_ID	Payment transactions and payment status
subscriptions	Subscription_ID	Plan, monthly fee, contract and support tier

support_tickets	Ticket_ID	Support contacts, resolution and satisfaction
monthly_usage	Usage_ID	Monthly engagement, sessions, data usage and usage issues

Customer_ID is unique in customers and is the logical foreign key in payments, subscriptions, support_tickets and monthly_usage.

3. Data Cleaning & Quality Assurance

Data was profiled in Python and validated in MySQL before dashboarding. Important cleaning decisions were documented rather than silently changing values.

Issue	Action / decision
Customer_ID stored as text	Kept as VARCHAR because identifiers such as CUST00001 are not numeric measures.
20 duplicate payment records	Removed exact duplicate payment rows; payments reduced from 117,521 to 117,501.
Blank Churn_Date for active customers	Converted blank strings to true NULL values; 10,394 active customers have NULL Churn_Date and 1,606
Missing Satisfaction_Score	35 values retained as legitimate NULLs rather than inventing scores.
Primary keys	Added and validated primary keys for all five tables.
Dates and numeric fields	Converted/validated dates and checked important numeric/category fields.

Join-safety principle

Payments, support tickets and monthly usage are one-to-many tables. Raw joins across several child tables can multiply rows and inflate totals. For customer-level metrics, child tables should be aggregated by Customer_ID before combining multiple one-to-many sources.

4. Python Analysis

Python/pandas was used for data understanding, cleaning, validation and exploratory analysis: profiling, null/duplicate checks, datetime conversion, ID validation, payment deduplication, churn metrics, segmentation, customer-level payment value, and support/usage analysis.

5. SQL Analysis

MySQL was used to independently answer the business questions. The workflow was business question → tables/columns → filters → JOIN → aggregation → GROUP BY → validation.

Analysis area	Examples
Churn	Overall churn; region/state/city; monthly trend; plan and contract
Acquisition	Acquisition channel and primary device
Customer value	Successful revenue; customer value vs churn; payment amount by status
Payments	Revenue by method; failed amount; payment failure rate
Support	Average resolution; average satisfaction; satisfaction groups
Usage	Average usage hours; average usage issue count

6. Tableau Dashboard Design

The final workbook contains three dashboards arranged as a business story: **Executive Overview** → **Churn Drivers** → **Customer & Revenue Impact**.

Dashboard 1 — Executive Churn Overview

Focus: how large is the churn problem? Includes total customers, churn rate, active vs churned customers, monthly churn trend, churn by contract type, region and plan.

Dashboard 2 — Churn Drivers

Focus: which customer, payment, support and usage dimensions are associated with higher churn? Includes acquisition channel, primary device, payment failure rate, tenure, support, satisfaction, usage and a Plan × Contract Type heatmap.

Dashboard 3 — Customer & Revenue Impact

Focus: where is business exposure concentrated? Includes historical successful payment value associated with churned customers, plan-level revenue exposure, acquisition-channel performance and top cities by churned customers.

7. Key Observations

These are descriptive associations in a fictional dataset and should not be interpreted as causal findings.

Observation	Evidence
Overall churn	1,606 of 12,000 customers are churned, giving a 13.38% overall churn rate.
Monthly peak	May 2026 recorded 477 churned customers; March 366, April 383 and June 380.
Contract type	Monthly contracts show 17.45% churn, versus 11.14% quarterly and 6.54% annual.
Subscription plan	Basic shows the highest observed plan churn at 15.65%.

Acquisition channel	Partner shows the highest observed channel churn at 14.69%.
Primary device	Web shows the highest observed device churn at 14.28%.
Revenue exposure	Historical successful payment value associated with churned customers totals ■23.72M.

8. Business Recommendations

Prioritize monthly-contract retention

Evidence: Monthly contracts have the highest observed churn rate. **Action:** Review onboarding, renewal friction, pricing and retention offers for monthly customers.

Investigate the Basic plan experience

Evidence: Basic has the highest observed plan churn. **Action:** Compare value perception, feature usage and service interactions for Basic customers.

Audit Partner acquisition

Evidence: Partner acquisition has the highest observed channel churn. **Action:** Review partner onboarding, customer expectations and early-life service quality.

Investigate the May churn spike

Evidence: May has the highest monthly churn count. **Action:** Review product, billing, service and operational events around May.

Protect high-value churn exposure

Evidence: Churned customers are associated with █23.72M of historical successful payment value. **Action:** Prioritize retention analysis for high-value plans and customers.

Use rate and count together

Evidence: Churn rate supports fair segment comparison while count/value shows absolute impact. **Action:** Use both measures when setting retention priorities.

9. Portfolio & Technical Deliverables

Recommended repository contents: processed CSV files; Python notebooks; SQL scripts; Tableau workbook or public-workbook link; dashboard screenshots; and this documentation PDF.

10. Skills Demonstrated

Python: pandas, profiling, cleaning, datetime handling, grouping, aggregation and validation.

SQL: SELECT, WHERE, GROUP BY, ORDER BY, CASE, COUNT, SUM, AVG, JOINS, subqueries/CTEs and validation.

Tableau: calculated fields, KPIs, filters, trend analysis, churn analysis, heatmaps and dashboard storytelling.

Analytics: churn, segmentation, customer value, payment behavior, support and usage analysis.

11. Limitations

PulseNet is a fictional dataset. The analysis is observational and descriptive. Higher churn in a segment does not prove causation. Historical successful payment value associated with churned customers should not be described as guaranteed future revenue loss or recurring revenue without a suitable revenue model.

12. Final QA Status

The project has completed the intended workflow: data understanding, cleaning, SQL validation, Python analysis, Tableau dashboard creation and portfolio documentation. It is designed as a beginner-level Data Analyst portfolio project with an emphasis on clear business questions and practical analytics.